

MATH 3115 OMS & OED COMPLEX ANALYSIS

Lecture 3: Powers and Roots of complex numbers

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Lesson Outcome

By the end of this lesson you should be able to:

- Solve powers of complex numbers.
- Apply Trigonometric identities in solving complex numbers .
- Use binomial expansion to solve complex numbers
- Use De Moivre formula to solve complex numbers

INTRODUCTION

- So far, we know that we can multiply two complex numbers in Cartesian form (in the form of $(a + bi)$).
- We have also learned that we can multiply two complex numbers in polar form.
- When we multiply powers of complex numbers, the algebraic computations become tedious.
- In this lecture we will multiply powers of complex numbers in polar form.

Complex numbers in Polar form

Example 1

Find the product of $(1 + i)(1 + i)$.

Solution 2

$$\begin{aligned}(1 + i)(1 + i) &= 1 + i + i + i^2 \\ &= 1 + 2i - 1 \\ &= 2i\end{aligned}$$

Complex numbers in Polar form CONT'D

Example 3

Find the product of

$$(1 + i)(1 + i)(1 + i)(1 + i) = (1 + i)^4 .$$

Solution 4

$$\begin{aligned}(1 + i)(1 + i)(1 + i)(1 + i) &= (1 + i)^2(1 + i)^2 \\ &= (2i)(2i) \\ &= 4i^2 \\ &= 4(-1) = -4\end{aligned}$$

Example 5

Let $z = (1 + i)$, find $(1 + i)^2$ in polar form.

Solution to example 5 CONT'D

Solution 6

$$r = \sqrt{1^2 + 1^2} = \sqrt{1 + 1} = \sqrt{2}$$

$$\theta = \tan^{-1}(1/1) = \tan^{-1}(1) = \pi/4$$

$$z = \sqrt{2} \left(\cos \pi/4 + i \sin \frac{\pi}{4} \right)$$

$$\begin{aligned} \text{Therefore } z^2 &= (1 + i)^2 = \left(\sqrt{2} \left(\cos \pi/4 + i \sin \frac{\pi}{4} \right) \right)^2 \\ &= (\sqrt{2})^2 (\cos \pi/4 + i \sin \pi/4)^2 \\ &= 2(\cos \pi/4 + i \sin \pi/4)(\cos \pi/4 + i \sin \pi/4) \\ &= 2(\cos(\pi/4 + \pi/4) + i \sin(\pi/4 + \pi/4)) \\ &= 2(\cos 2\pi/4 + i \sin 2\pi/4) \text{ using trig identity} \\ &= 2 \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \right) \\ &= 2(0 + i) \\ &= 0 + 2i \\ &= 2i \end{aligned}$$

Trigonometric identities in complex numbers

$$\begin{aligned} z^2 &= [r(\cos \theta + i \sin \theta)]^2 \\ &= r^2(\cos 2\theta + i \sin 2\theta) \end{aligned}$$

This comes directly from the addition formula of trigonometric identities.

$$\cos(\theta + \phi) = \cos \theta \cos \phi - \sin \theta \sin \phi$$

$$\sin(\theta + \phi) = \sin \theta \cos \phi + \cos \theta \sin \phi$$

If $\theta = \phi$ like in the case of a double angle, the formulas become

$$\cos(\theta + \theta) = \cos 2\theta = \cos \theta \cos \theta - \sin \theta \sin \theta$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

● This simplifies evaluation of complex numbers.

De Moivre CONT'D

Example 7

Simplify $(1 - \sqrt{3}i)^6$

- We first express the number in polar form.
- Then use the trigonometric formula to evaluate.
- Lets see the solution in the next slide.

De Moivre CONT'D

Solution 8

$$r = \sqrt{1^2 + (-\sqrt{3})^2} = \sqrt{1+3} = 2$$

$$\theta = \tan^{-1} \left(-\frac{\sqrt{3}}{1} \right) = \tan^{-1}(-\sqrt{3}) = -\tan^{-1}(\sqrt{3}) = -\frac{\pi}{3}$$

since the number is in the fourth quadrant, $\theta = \frac{5\pi}{3}$

$$\begin{aligned}(1 - \sqrt{3}i)^6 &= \left[2 \left(\cos \frac{5\pi}{3} + i \sin \frac{5\pi}{3} \right) \right]^6 \\&= 2^6 \left(\cos 6 \left(\frac{5\pi}{3} \right) + i \sin 6 \left(\frac{5\pi}{3} \right) \right) \text{ by De Moivre's Formula} \\&= 2^6 (\cos 10\pi + i \sin 10\pi) \\&= 64(-1 + 0i) \\&= -64\end{aligned}$$

Demoivre formular

- In general, we have

$$z^2 = [r(\cos \theta + i \sin \theta)]^2$$

$$= r^2(\cos 2\theta + i \sin 2\theta)$$

$$z^3 = r^3(\cos 3\theta + i \sin 3\theta)$$

$$\vdots$$

$$z^n = r^n(\cos n\theta + i \sin n\theta) \quad \forall n \in \mathbb{Z}$$

- when $r = 1$, we have

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta.$$

- This is known as De Moivre's Formula.

Abraham De Moire CONT'D

Theorem 9 (Abraham De Moivre)

For every real number θ and every positive integer n , we have

$$\cos(n\theta) + i \sin(n\theta) = (\cos \theta + i \sin \theta)^n.$$

- De Moivre's Theorem is an important identity useful in finding powers and roots of complex numbers.
- It also helps in deriving trigonometric identities for $\cos(n\theta)$ and $\sin(n\theta)$.

Abraham De Moivre CONT'D

Example 10

Example Find the value of $(\cos 3^\circ + i \sin 3^\circ)^{40}$.

Solution 11

By De Moivres Theorem

$$\begin{aligned}(\cos 3^\circ + i \sin 3^\circ)^{40} &= (\cos 3 \times 40 + i \sin 3^\circ \times 40) \\&= \cos 120^\circ + i \sin 120^\circ \\&= \frac{1}{2} + \frac{\sqrt{3}}{2}i\end{aligned}$$

Corollary 12

For every real number we have

$$(a) \cos(2\theta) = \cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta - 1$$
$$\sin(2\theta) = 2 \sin \theta \cos \theta.$$

$$(b) \cos(3\theta) = \cos^3 \theta - 3 \cos \theta \sin^2 \theta = 4 \cos^3 \theta - 3 \cos \theta$$
$$\sin(3\theta) = 3 \cos^2 \theta \sin \theta - \sin^3 \theta$$

The proof for the corollary is left for the reader as an exercise.

Example 13

Show that $(1 + i)^5 = -4 - 4i$.

Solution 14

$$r = \sqrt{2}$$

$$\theta = \tan^{-1}(1/1) = \tan^{-1}(1) = \pi/4$$

$$z = 1 + i = \sqrt{2}(\cos \pi/4 + i \sin \pi/4)$$

$$z^5 = (1 + i)^5 = \left(\sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right) \right)^5$$

$$= (\sqrt{2})^5 \left(\cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} \right) \text{ *Demoivre*}$$

$$= 4\sqrt{2} \left(-\frac{1}{\sqrt{2}} + i \left(\frac{-1}{\sqrt{2}} \right) \right)$$

$$= 4\sqrt{2} \left(-\frac{1}{\sqrt{2}} - i \frac{1}{\sqrt{2}} \right)$$

$$= -4 - 4i$$

Binomial expansion in complex numbers

Example 15

Show that $\cos 5\theta = \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta$

Solution 16

$\cos 5\theta = \operatorname{Re}(\cos 5\theta + i \sin 5\theta)$.

Hence $(\cos 5\theta + i \sin 5\theta) = (\cos \theta + i \sin \theta)^5$ by De Moivre

Taking $n = 5$, and using Binomial Formula

$$\begin{aligned}(\cos 5\theta + i \sin 5\theta) &= \binom{5}{0} \cos^5 \theta (i \sin \theta)^0 + \binom{5}{1} \cos^4 \theta (i \sin \theta)^1 + \binom{5}{2} \cos^3 \theta (i \sin \theta)^2 \\&+ \binom{5}{3} \cos^2 \theta (i \sin \theta)^3 + \binom{5}{4} \cos^1 \theta (i \sin \theta)^4 + \binom{5}{5} \cos^0 \theta (i \sin \theta)^5 \\&= \cos^5 \theta + 5i \sin \theta \cos^4 \theta + 10i^2 \cos^3 \theta \sin^2 \theta + 10 \cos^2 \theta \sin^3 \theta i^3 + 5 \cos \theta i^4 \sin^4 \theta + i^5 \sin^5 \theta \\&= \cos^5 \theta + 5i \cos^4 \theta \sin \theta - 10 \cos^3 \theta \sin^2 \theta - 10i \cos^2 \theta \sin^3 \theta + 5 \cos \theta \sin^4 \theta + i \sin^5 \theta \\&= \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta + i (5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta)\end{aligned}$$

Equating the real and imaginary parts, we get

$$\cos 5\theta = \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta.$$

Roots of complex numbers

Definition 17

The n^{th} power of a complex number z is given by $z^n = [r(\cos \theta + i \sin \theta)]^n$. That is $z^n = r^n(\cos n\theta + i \sin n\theta)$, $n \in \mathbb{N}$, by using De Moivre's theorem.

- We can find the n^{th} root of the complex number $z = r(\cos \theta + i \sin \theta)$. It follows

The n^{th} root of the complex number

Definition 18

If

$$\begin{aligned}w^n &= z \\w &= z^{1/n} = (r(\cos \theta + i \sin \theta))^{1/n} \\&= r^{1/n} \left(\cos \frac{\theta}{n} + i \sin \frac{\theta}{n} \right)\end{aligned}$$

Since cosine and sine are 2π periodic, then we can rewrite

$$\begin{aligned}w &= r^{1/n} \left(\cos \left(\frac{\theta + 2\pi k}{n} \right) + i \sin \left(\frac{\theta + 2\pi k}{n} \right) \right) \\&\text{for } k = 0, 1, 2, \dots, m-1\end{aligned}$$

Example 19

Find the cube roots of $8i$.

Solution 20

$$z = 0 + 8i$$

$$r = \sqrt{0^2 + 8^2} = \sqrt{8^2} = 8$$

$$\theta = \tan^{-1}(8/0) = \tan^{-1}(\infty) = \pi/2$$

*So, we have $8i = 8 \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \right)$.
and since cosine and sin are 2π periodic, Therefore we
get*

Solution 20 to example 19 CONT'D

Solution 21

$$w_k = 8^{1/3} \left(\cos \frac{\pi/2 + 2\pi k}{3} + i \sin \frac{\pi/2 + 2\pi k}{3} \right), k = 0, 1, 2$$

$$k = 0, \quad w_0 = 2 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right) = 2 \left(\frac{\sqrt{3}}{2} + i \frac{1}{2} \right) = \sqrt{3} + i$$

$$\begin{aligned} k = 1, \quad w_1 &= 2 \left(\cos \frac{\pi/2 + 2\pi}{3} + i \sin \frac{\pi/2 + 2\pi}{3} \right) \\ &= 2 \left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right) = 2 \left(-\frac{\sqrt{3}}{2} + i \frac{1}{2} \right) = -\sqrt{3} + i \end{aligned}$$

$$k = 2, \quad w_2 = 2 \left(\cos \frac{\pi/2 + 4\pi}{3} + i \sin \frac{\pi/2 + 4\pi}{3} \right)$$

$$\begin{aligned} k = 2, \quad w_2 &= 2 \left(\cos \frac{\pi/2 + 4\pi}{3} + i \sin \frac{\pi/2 + 4\pi}{3} \right) \\ &= 2 \left(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2} \right) \text{ from unit chart } (0, -1) \\ &= -2i \end{aligned}$$

THE END